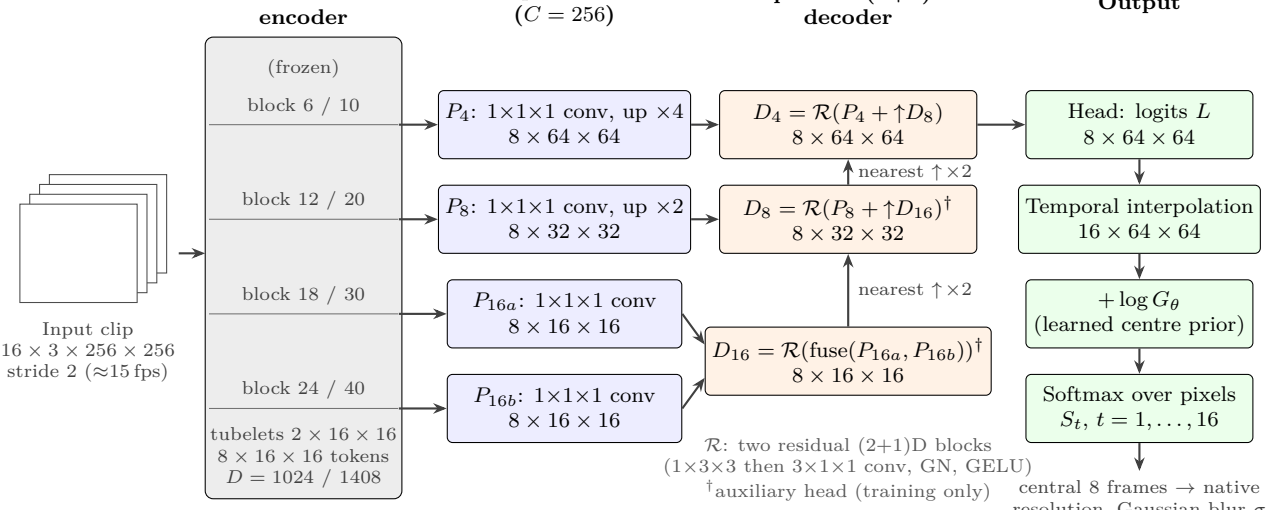

V-JEPA 2 encoder
SimpleFPN-3D ($C = 256$)
Top-down (2+1)D decoder
Output
(frozen)
block 6 / 10
block 12 / 20
block 18 / 30
block 24 / 40
tubelets 2 × 16 × 16
8 × 16 × 16 tokens
$D = 1024$ / 1408
Input clip
16 × 3 × 256 × 256
stride 2 (≈15 fps)
P_4: 1×1×1 conv, up ×4
8 × 64 × 64
P_8: 1×1×1 conv, up ×2
8 × 32 × 32
P_{16a}: 1×1×1 conv
8 × 16 × 16
P_{16b}: 1×1×1 conv
8 × 16 × 16
$D_4 = \mathcal{R}(P_4 + \uparrow D_8)$
8 × 64 × 64
nearest ↑ ×2
$D_8 = \mathcal{R}(P_8 + \uparrow D_{16})^\dagger$
8 × 32 × 32
nearest ↑ ×2
$D_{16} = \mathcal{R}(\text{fuse}(P_{16a}, P_{16b}))^\dagger$
8 × 16 × 16
$\mathcal{R}$: two residual (2+1)D blocks
(1×3×3 then 3×1×1 conv, GN, GELU)
†auxiliary head (training only)
Head: logits L
8 × 64 × 64
Temporal interpolation
16 × 64 × 64
$+ \log G_\theta$
(learned centre prior)
Softmax over pixels
S_t, $t = 1, \ldots, 16$
central 8 frames → native
resolution, Gaussian blur σ